Why These Simple Laws, in Brief

Gary Abraham Bernstein

Independent Researcher ORCID: https://orcid.org/0009-0009-1761-2867

These versions share material and belong to overlapping submission families.

Matheism defends identity between reality and complete mathematical structure. A mathematical description alone does not establish that identity: the argument also requires structural exhaustion, the claim that no constitutive fact remains outside the complete structure. The use of structured reasoning by an opponent is not itself a proof of that premise.

Induction can be studied using explicit priors over computable models. Convergence results have conditions; short laws can themselves prescribe changes in time. A general proof that unchanging laws always have the shortest description does not follow.

Abstract

A universe need not list every particle if a short rule generates them. Under the stated simplicity measure, saving five bits gives a description 32 times the initial weight. The saving must survive a complete accounting of laws, starting conditions and predicted observations. This paper applies that principle to quantum theory and observer development: avoid independently specifying what a common rule can explain. If a history begins without the organization needed for its later observers, that organization must develop. This necessary development does not require the laws themselves to become more complex. The proposal is to explain our intricate universe through a short complete account; identifying the account and its observational weight is the remaining physical task.

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1. One Principle

How much must we specify to explain the world we observe? The answer includes the laws, the starting conditions and how observers obtain their records. A short equation with an elaborate unexplained starting state may be a long explanation in disguise. The aim is to count the whole account once.

Here the physical work is to recover geometry, quantum statistics and the organization of observers within economical candidates. The following comparison rule makes that search quantitative.

2. A Quantified Simplicity Preference

Fix a description language and a class of complete candidate models. With assigned code weights w(H) proportional to 2^(−L(H)), a genuine difference of ΔL bits gives a prior ratio 2^ΔL. Ten bits gives exactly 1,024. Posterior odds additionally multiply by the evidence likelihood ratio. Summing over all programs producing an output is different from assigning only its shortest program weight. Neither construction automatically defines a normalized measure over every mathematical structure or every observer.

For fixed candidate descriptions with equal likelihoods, posterior odds inherit their prior ratio. Unequal predictions, alternative code choices and multiplicities can change the comparison.

Historical unifications motivate searching for economical predictive theories. They do not establish a universal measured reduction in Kolmogorov complexity.

3. The Principle of Least Action

Classical mechanics, general relativity, quantum field theory, and the Standard Model all admit variational formulations. Systems evolve along paths extremizing the action integral. Why do the fundamental theories all admit variational formulations?

Compare complete generators of the same observations in one description language. An action formulation may be economical, but comparing it only with an exhaustive trajectory list does not establish that all nonvariational laws are expensive. Noether's theorem supplies a relation within variational theories, not a proof that any possible observer requires one.

Noether's theorem relates symmetries and conservation laws in a variational setting. Symmetry can compress some models; maximal symmetry is not a universal prediction of simplicity or a description of all observed physics.

The search compares generators reproducing the relevant observables. The commonly cited Standard Model parameter count is not a count of particles. Model content, precision and neutrino conventions must be fixed before estimating description length.

4. Fine Tuning as a Model Comparison

Sensitivity of observers to constants is model dependent. Claims about which variations prevent stars, nuclei or chemistry require a defined family of variations.

Universal realization is a further ontological commitment. If every consistent structure is physically realized, consistent selection rules are included. The reverse inference does not follow: the consistency of a filter, including the filter

that excludes nothing, does not establish its physical realization without assuming the ensemble being argued for. Mathematical monism motivates the proposal but does not by itself prove that every mathematical possibility is physical.

An observer conditioned simplicity measure could constrain constants once the candidate family and observer weights are specified. No code independent bit ranking of pi, e or other constants is used here.

5. The Arrow of Time

The proposed arrow concerns the production of information. In a history with a first event and no last, forward generation can begin with supplied information. A reverse dependency chain in which every step waits for a later step cannot start without an independently supplied input. This excludes that reverse construction. Connecting it to experienced time requires showing why experience needs such production, rather than the relevant relations simply existing together in a mathematical block.

If a history starts without the organization its later observers require, that organization must develop. The laws themselves can remain simple. In biological histories like ours, available energy supports that development while total entropy increases. Neither organization nor entropy must increase at every step.

Coordinated convergences in reversed splashes may depend on details a local learner cannot sense. If the same available evidence hides situations requiring incompatible responses, no decision based on that evidence can guarantee success in both. Later observations may explain the event, too late for the earlier choice. Repeated dependence on such inaccessible information could constrain the evolution of reliable learners. A directional preference must survive a comparison of observers and environments together: full reversal also transforms the records.

Start, hypothetically, with neither temporal direction favored: a coin flip. The proposed arguments then favor forward development through rooted information production, the development of initially absent observer organization and reliable evidence available before action. Even if reversal is possible, it could receive less probability weight under the stated measure. Establishing that preference requires showing that additional specification or restrictions on viable observer histories actually reduce their total weight, including comparison with a fully reversed observer and environment. The arguments share assumptions and cannot be counted as independent probability gains.

Reversing a real billiard break requires scattered light, sound and heat to return in coordination. If specifying that coordination costs k extra bits, the reverse description receives $2^{\wedge}(-k)$ times the weight: every extra bit halves it. A short common rule could specify those arrivals, so the additional cost must be established for the complete descriptions. Appending a reversal instruction to one

program does not prove that every reverse program is longer.

6. The Case for Common Generation

Retain local responses, definite outcomes and the requirement to recover Bell violations. Within the stated Bell framework, measurement independence must fail. Superdeterminism explains the dependence by placing source, apparatus and setting devices inside a common physical generator. This is the local route investigated here.

Specificity is not complexity. The digits of pi are fixed by a compact rule. Likewise, a precise relation among measurement settings and hidden data need not be entered as a separate instruction for every event. Pseudorandom settings remain physical processes. Their irregular appearance establishes neither independence from the measured system nor fundamental chance.

Alice's and Bob's settings can be mutually independent while their pair is correlated with the hidden data. Recovering the quantum value and its full dependence on preparation is the generator's physical task.

Compare joint descriptions rather than adding unconditional law, input and outcome complexities as though they shared no information. In predictive comparisons, observed records enter through likelihoods for every candidate. A short stochastic law need not list its outcomes, and an Everettian account must explain their observational weights. Under a fixed description prior, a verified shorter complete account retains its advantage when likelihoods agree.

The block view motivates the internal treatment of experimental settings. The Bell constraint, together with the retained locality and outcome premises, establishes the required dependence. Those are complementary parts of the argument.

7. Beyond Physics

The position extends beyond physics: consciousness, AI alignment, and ethics follow as structural consequences and are developed in companion papers.

8. A Connected Search

The same discipline links the applications: state the generator, charge its boundary, recover the observations and compare the complete account. The strongest next result would do more than fit one puzzle. It would use the same compact structure to explain quantum correlations, effective geometry and the conditions for observers.

A simplicity prior makes comparative predictions once the candidate family and likelihoods are supplied. Evidence can reverse a prior preference. A verified description saving with equal likelihood preserves it. This provides a concrete

search rule without requiring a claim that every future law must be shorter than the last.

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The central proposal is to explain more physics with fewer independently specified ingredients. Under the declared simplicity rule, a measured saving in bits gives an exact advantage in initial weight. Observations then determine how well each candidate fits.

This approach connects quantum theory, time and observer development through a common accounting of laws, starting conditions and records. Its value comes from complete explanations that share structure.

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